

# Asymmetric interfaces with stylus and gesture for VR sketching

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## ABSTRACT

Virtual Reality (VR) can be used for design and artistic applications. However, traditional symmetrical input devices are not specifically designed as creative tools and may not fully meet artist needs. In this demonstration, we present a variety of tool-based asymmetric VR interfaces to assist artists to create artwork with better performance and easier effort. These interaction methods allow artists to hold different tools in their hands, such as wearing a data glove on the left hand and holding a stylus in the right-hand. We demonstrate this by showing a stylus and glove based sketching interface. We conducted a pilot study showing that most users prefer to create art with different tools in both hands.

**Index Terms:** Human-centered computing—Virtual Reality—Asymmetric Interface

## 1 INTRODUCTION

Artists often use two hands to operate different input tools to create their artwork. This tool-based asymmetric interaction design has been carried over from traditional art workflow to digital art creation. For example, the combination of a digital tablet and keyboard or mouse and keyboard. In contrast, most mainstream Virtual Reality (VR) devices use symmetric controllers such as two handheld controllers that are the same or a two-handed gesture interface.

In our research, we are interested in exploring asymmetric VR interfaces and how these might be used to improve the artistic process. This demonstration integrates data glove and stylus input into a tool-based asymmetric sketching application. We have also conducted a pilot study to compare it with three different bimanual input solutions. Figure 1 presents the initial configuration of the interface.

## 2 RELATED WORK

Previous research found that the stylus input was more effective for precise input in VR [4, 5]. Additionally, several studies proposed methods such as providing pressure sensing [10], physical feedback [1], and improved grip posture to optimize the user experience [9].

The motivation of this work is to enhance the user experience of sketching in VR by making it similar to the user's real-world working habits, in particular, using different tools with both hands. The value of asymmetric bi-manual input for optimizing the user experience has been presented in many studies [2, 3, 7]. These studies mention the need for asymmetric interfaces to enhance the user experience in certain VR tasks. In this research, we explore the impact on the performance and usability of VR sketching by integrating stylus and gesture input into an asymmetric interface.

## 3 SYSTEM DESIGN

Inspired by our previous work [11], we used three devices for input in our VR system: two standard HTC Vive controllers, a StretchSense

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Figure 1: Our asymmetric VR Sketching interface.

data glove with real-time hand gesture detection, and the Logitech VR Ink [6] stylus with the 6 degrees of freedom (6 DoF) tracking:

- The HTC VIVE controller is used the same way and functions as in the standard case. The user could pull the right trigger to draw strokes and the left trigger to grab the item.
- The Logitech VR Ink's pressure-sensitive button on the stylus allows the user to intuitively adjust the thickness of the generated strokes. The stylus is also held in a grip closer to a realistic pen grip.
- The StretchSense [8] data glove provides gesture-based input, allowing the user to use pinch gestures to select, move and rotate objects. An additional add-on VIVE tracker provides spatial location information for the data glove.

We developed our test application using the open-source Tilt Brush software<sup>1</sup>, which was running on a desktop computer with an HTC VIVE VR headset tethered to it.

Our input system allows users to use four input modes for VR sketching, including the traditional dual HTC Vive controller, left-hand Vive controller with right-hand stylus, left-hand data glove with right-hand stylus, and left-hand data glove with right-hand Vive controller. These four input modes were designed to explore different potential needs in VR sketching tasks. Figure 3 shows the interface metaphors that we developed. All four input methods were designed to allow the user to use the right-handed tool to create strokes and the left-handed tool to move and rotate to select objects. Figure 2 shows the screenshot of each method.

## 4 DEMONSTRATION

In this demonstration, we will live stream into Zoom a VR view of a person sketching using the four different interface configurations.

<sup>1</sup><https://github.com/icosa-gallery/open-brush>

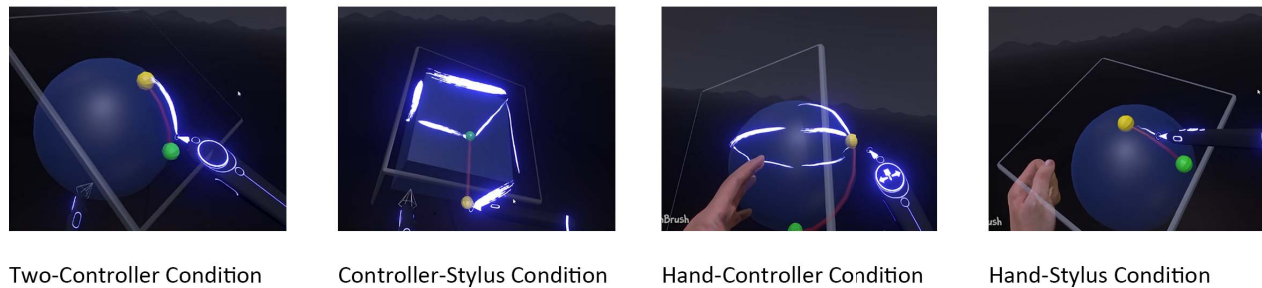


Figure 2: Screenshots of our four interaction methods running in VR.

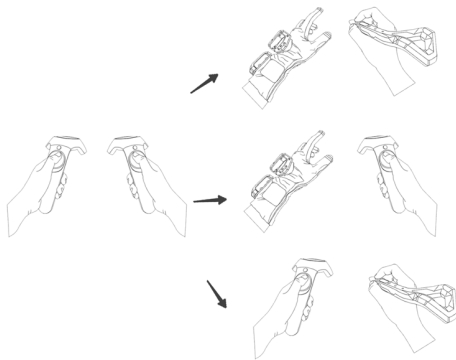


Figure 3: The design of our interaction methods.

The user has to draw straight lines, curves, and complex spatial curves using the given semi-transparent model as a base using different interaction methods. We used the preset assets from Open Brush as the virtual interface in VR, and the hand models for gesture input were taken from the StretchSense preset assets. The viewers will be able to ask questions of the user and get their feedback on the advantages and disadvantages of each interface combination. We will also have some videos and presentation material to show, including the results of a pilot study conducted with the system (see below).

## 5 PILOT STUDY AND RESULT

We conducted a pilot study with 8 subjects to study the usability of our four developed interfaces using a within group design. We collected subjective feedback through the System Usability Scale (SUS) questionnaire. We also interviewed the participants after the experiment.

A Friedman test compared the usability of each method. Overall, the two-controller condition had the highest usability score ( $M=83.8$ ,  $SD=16.3$ ), the controller-stylus condition had a good usability score ( $M=75.0$ ,  $SD=14.9$ ). In contrast the hand-controller and hand-stylus condition's usability score remained the same ( $M=65.0$ ,  $SD=13.9$ ). The post-study questionnaire showed that the controller-stylus condition had the highest user preference, followed by the two-controller condition, hand-stylus condition, and hand-controller condition, respectively.

## 6 CONCLUSION AND DISCUSSION

Our pilot study suggested that asymmetric interaction with stylus input and gesture input has the potential to be a good VR sketching

interface, with a higher usability score than the other tested alternatives. The precise and natural operation of the stylus input can be combined with the intuitive gesture input and then shape a good user experience. Based on verbal feedback and pilot study results, we infer that the reason for the lower interaction usability and user preference for the two gesture input condition is that the tracking of the data glove is not yet stable enough, which may increase the user error rate during the operation. In addition, since the participants have extensive VR experience, they are familiar with the standard HTC VIVE controller, which may affect the experimental results to some extent. In the future, we plan to improve the data glove input accuracy, optimize the gesture input performance, and recruit participants with design backgrounds to conduct formal user studies.

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